Student Worksheet 5.6A

Extra Practice Questions: Applications of Stoichiometry

1. In a chemical analysis to test the purity of a bottle of sodium bromide, a solution containing 1.17 g of sodium bromide was reacted with an excess of dimercury(I) acetate solution. The dry precipitate had a mass of 2.73 g. Calculate the percent yield for the precipitate and comment on the purity of sodium bromide.

2. A solution containing 2.56 g of aluminum nitrate is mixed with a solution containing 1.02 g of ammonium sulfide. Determine the unreacted mass of the excess reagent and the mass of precipitate formed.